Photon Ontology and Fold Kinematics in a Phase-Transitioning Vacuum

Propagation, Momentum Transfer, and the Relativistic Dispersion Relation as Stability Budget Constraints

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Abstract

Within the Dynamic Principles of the Substrata (DPS) framework, the vacuum is a real, continuous, hyper-elastic medium capable of phase transitions. This paper develops the ontology of the photon as a propagating phase-transition event and derives, from a single stability condition, the following results: (i) the photon's velocity c is an existence condition, not a speed limit; (ii) $E \propto f$ emerges mechanically from substrate compression dynamics; (iii) photon amplitude is bounded by the phase-transition threshold, predicting that radiation intensity is photon flux rather than oscillation amplitude; (iv) momentum transfer during photon absorption follows directly from stability-budget reallocation; (v) the relativistic energy–momentum relation $E^2 = (pc)^2 + (m_0c^2)^2$ is argued to follow from orthogonality of the stability budget's compression and traversal channels; and (vi) the impossibility of massive particles reaching c is a topological closure requirement. These results require no new free parameters and generate falsifiable predictions including a substrate-derived Schwinger limit and frequency-dependent Compton kinematics.

Keywords: photon ontology, phase transition, vacuum as medium, emergent relativity, energy–momentum relation, fold kinematics, Compton scattering, Schwinger limit, hyper-elastic continuum, topological defects, substrate model

1 Introduction

The photon occupies a peculiar position in modern physics. It is simultaneously a quantum of the electromagnetic field, a massless boson, a carrier of momentum despite having no rest mass, and an entity whose wave–particle duality remains a subject of interpretive debate a century after its identification. These properties are not derived from a common mechanical principle; they are separately postulated or extracted from independent theoretical structures [4, 2, 1].

The Dynamic Principles of the Substrata (DPS) framework [13] treats the vacuum as a real, continuous, hyper-elastic medium capable of phase transitions between vacuum, dark, and baryonic phases. Within this framework, particles are metastable topological folds of the substrate, and the propagation of light is a phase-transition phenomenon rather than a wave in an elastic medium. This approach shares conceptual ground with analog gravity programmes [10, 11] and emergent spacetime models [12], but differs in treating the substrate as a phase-transitioning solid rather than a superfluid or condensate.

Previous work within DPS has demonstrated that this framework recovers gravitational lensing as anisotropic refraction [14], Lorentz invariance via substrate stress feedback [15], Shapiro delay as constitutive retardation [16], and galactic rotation curves from substrate mode scarcity [17], all from constitutive substrate properties with-

out invoking spacetime curvature or dark matter particles.

This paper develops the photon ontology in full and extends it to a general theory of fold kinematics.

1.1 Scope and Non-Claims

This paper derives consequences of a single stability condition applied to substrate folds. It does *not* attempt a full quantum measurement theory, a derivation of the Born rule, or a complete account of entanglement. The stability condition itself is postulated on physical grounds and motivated by the substrate's constitutive mechanics; its formal derivation from the DPS action principle is the subject of ongoing work. Where derivations in this paper rely on physical arguments rather than formal proofs, this is stated explicitly.

Section 2 introduces the stability condition governing all substrate folds. Section 2.3 defines notation and dimensions. Section 3 derives photon propagation, frequency, amplitude bounds, and polarization. Section 4 derives momentum transfer, the relativistic energy–momentum relation, and the massive-particle speed limit. Section 5 identifies falsifiable predictions.

2 The Stability Condition

2.1 Phase Transitions in the Substrate

The substrate exists locally in one of three phases: vacuum (solid-like, supporting wave propagation), dark (amorphous, pressure-bearing, electromagnetically inert), and baryonic (topologically folded, locked). The boundaries between these phases are characterized by energy thresholds.

When a region of vacuum substrate receives a perturbation of sufficient energy, it undergoes a local phase transition toward the baryonic phase. Two outcomes are possible:

(i) If the perturbation energy meets or exceeds the *stability threshold* q_{stable}, a stable topological fold forms—a baryonic particle with rest mass.

(ii) If the perturbation exceeds the *phase-transition threshold* but remains below q_{stable}, an unstable fold nucleates and must propagate to persist. This is a photon.

2.2 The Master Equation

Every substrate fold—stable or unstable—must satisfy the instantaneous stability condition:

$$q_c + \frac{q_t}{\Delta t} = q_{\text{stable}} \tag{1}$$

The variables are defined in Table 1.

This equation is the central object of the paper. It governs photon propagation, particle rest, momentum transfer, and relativistic kinematics from a single balance relation.

2.3 Notation and Dimensions

3 Photon as Propagating Phase Transition

3.1 Existence at c

For a photon, the compressed fold energy q_c is below the stability threshold: $q_c < q_{\text{stable}}$. The deficit must be compensated entirely by traversal:

$$\frac{q_t}{\Delta t} = q_{\text{stable}} - q_c > 0 \tag{2}$$

The fold must propagate through the substrate to satisfy Eq. (1) at every instant. The maximum traversal rate is c—the substrate's maximum phase-transition propagation velocity. Since the fold is sub-stable, it requires maximum traversal to persist; any velocity less than c would leave the stability budget in deficit, and the fold would collapse.

The photon does not travel at c because c is a speed limit. It travels at c because c is its *existence condition*. Reducing the traversal rate violates Eq. (1) and the fold ceases to exist.

3.2 Frequency as Oscillation Count

Energy above the minimum required to initiate the phase transition has no available path forward along the propagation vector—the fold is

Table 1: Notation, dimensions, and physical interpretation of stability-budget variables. The substrate displacement field $\mathbf{u}(\mathbf{x}, t)$, shear modulus μ, and density ρ are the constitutive variables of the DPS framework [13]. The precise functional mapping $q_c = q_c(\mathbf{u}, \nabla\mathbf{u}, \mu, \rho)$ is the subject of ongoing work connecting the stability condition to the substrate action principle.

Symbol	Meaning	Dimensions	Physical proxy
q_c	Compression energy of the fold	Energy [J]	Integral of strain energy density over fold volume: ∼
q_t	Traversal energy quantum	Energy [J]	Energy carried by the fold's displacement of substra
$q_t/\Delta t$	Traversal rate	Power [W]	Rate of substrate traversal; proportional to fold velo
q_{stable}	Stability threshold	Energy [J]	Minimum total energy for topological closure; local
c	Maximum phase-transition propagation velocity	Velocity [m/s]	$c = \sqrt{\mu_{\text{eff}}/\rho}$ at the phase boundary; the substrate's maximum signal speed
μ	Substrate shear modulus	Pressure [Pa]	Effective elastic stiffness of the vacuum phase
ρ	Substrate density	kg/m^3	Effective inertial density of the vacuum phase
ϵ_0	Energy per oscillation cycle	Energy [J]	Phase-boundary oscillation cost; identified with h

already phase-saturated at its leading edge. This excess energy manifests as substrate compression oscillations perpendicular to the propagation vector.

Each additional quantum of perturbation energy above the phase-transition threshold produces one additional oscillation cycle. Photon frequency f is therefore the count of perpendicular oscillation cycles per unit time:

$$E_{\text{photon}} = E_{\text{threshold}} + n \cdot \epsilon_0 \tag{3}$$

where n is the number of oscillation cycles and ϵ_0 is the energy cost per cycle—a substrate constant. In the limit where $E_{\text{threshold}}$ is small relative to the total energy:

$$E \propto f \tag{4}$$

The proportionality constant is h, which in this framework is identified as the energy cost per phase-boundary oscillation cycle: a mechanical property of the substrate, not a fundamental constant of quantum theory. This identification is consistent with the Planck relation [1] but provides a constitutive-mechanical origin for the constant.

3.3 Amplitude Bound and Intensity

The perpendicular oscillations are self-bounding. The transverse compression is insufficient to initiate a secondary phase transition in the perpendicular direction, confining the oscillation magnitude within the phase-transition threshold. Photon amplitude is therefore *not* an independent variable—it is bounded above by the substrate's phase-transition energy density.

This has a direct observational consequence: radiation intensity cannot be increased by increasing oscillation amplitude (which is capped). Intensity is therefore photon flux—the number of photons per unit area per unit time. This prediction is consistent with the photoelectric effect [2] and single-photon detection experiments, and provides a mechanical explanation for the quantization of light that is typically axiomatized.

3.4 Polarization

Polarization states follow without additional postulates:

- **Linear polarization**: transverse oscillations confined to a single plane.
- **Circular polarization**: the oscillation plane rotates about the propagation axis as excess energy distributes across successive perpendicular planes.
- **Elliptical polarization**: intermediate case.

All polarization states are substrate compression patterns bounded by the phase-transition

threshold.

3.5 Resolution of Wave–Particle Duality

The photon in DPS is a single mechanical object: a propagating, sub-stable phase-transition fold with perpendicular compression oscillations. It is simultaneously:

- **Localized**: the fold occupies a finite region of substrate and propagates along a definite vector.
- **Oscillatory**: it exhibits periodic transverse compression as a structural consequence of excess energy.

There is no duality. The "wave" and "particle" descriptions are two observable aspects of one mechanical structure. The complementarity principle is replaced by a complete mechanical description.

4 Fold Kinematics: The Stability Budget

4.1 Particle at Rest

A stable topological fold (baryonic particle) at rest satisfies Eq. (1) with:

$$q_c = q_{\text{stable}}, \qquad \frac{q_t}{\Delta t} = 0 \tag{5}$$

The entire stability budget is allocated to internal compression. No traversal is required. The fold persists in place.

4.2 Photon Absorption and Momentum Transfer

When a stable fold absorbs a photon, the photon's energy enters the fold. However, the fold's internal compression is already at the stability threshold q_{stable}—the maximum compression the substrate permits for a stable fold at that location. The absorbed energy cannot be stored as additional compression.

The excess must be expressed as traversal. After absorption:

$$q_{\text{stable}} = q_c + \frac{q_t}{\Delta t} \tag{6}$$

where now $q_t/\Delta t = q_{\text{photon}}$. The photon's traversal quanta—which sustained the photon's propagation at c—become the particle's traversal quanta. The particle *must move*.

The photon's energy, which was expressed as traversal (propagation at c), transfers directly into the particle's traversal budget. This is **momentum conservation as substrate accounting**: traversal quanta do not vanish when one fold merges with another, consistent with the standard conservation law [5].

The particle's velocity v after absorption is determined by the fraction of the stability budget now allocated to traversal:

$$v = c \cdot \frac{q_{\text{photon}}}{q_{\text{stable}}} \tag{7}$$

Since $q_{\text{photon}} \ll q_{\text{stable}}$ for typical photon–particle interactions, $v \ll c$, consistent with observation.

4.3 The Relativistic Dispersion Relation

The stability condition partitions the total energy budget into two channels:

$$E_{\text{rest}} = q_c \cdot c^2 \quad \text{(compression energy)} \tag{8}$$

$$E_{\text{kinetic}} = \frac{q_t}{\Delta t} \cdot c^2 \quad \text{(traversal energy)} \tag{9}$$

The total energy is:

$$E_{\text{total}} = q_{\text{stable}} \cdot c^2 \tag{10}$$

and the stability condition requires:

$$E_{\text{rest}} + E_{\text{kinetic}} = E_{\text{total}} \tag{11}$$

Identifying $E_{\text{rest}} = m_0 c^2$ and $E_{\text{kinetic}} = pc$ (since traversal at rate v carries momentum p), the standard relativistic energy–momentum relation is:

$$E^2 = (pc)^2 + (m_0 c^2)^2 \tag{12}$$

On the quadratic form. The stability condition (Eq. 1) is a *linear* partition of the energy budget. The relativistic dispersion relation (Eq. 12) is *quadratic*. The physical argument for the quadratic form is that compression and traversal are orthogonal energy channels in the substrate's configuration space: compression acts on the fold's internal topology while traversal acts along the propagation vector. Orthogonal contributions to a conserved total add in quadrature, as in the Pythagorean composition of independent vector components.

This argument is physically motivated but not yet formally derived from the substrate action principle. A rigorous derivation—showing that the functional form of the action $\mathcal{L}(\mathbf{u}, \dot{\mathbf{u}}, \nabla\mathbf{u})$ enforces quadratic addition of the compression and traversal energy functionals—is deferred to a dedicated paper on the variational structure of fold stability. The present claim is that the stability budget *motivates* the quadratic form through orthogonality; it does not yet constitute a proof.

This is the principal open derivation in the current paper. If the orthogonality argument fails under formal analysis, the stability-budget framework would still produce the linear results (Sections 3 and 4 up to this point) but would not recover the relativistic dispersion relation without modification.

4.4 Why Massive Particles Cannot Reach c

As a particle absorbs energy and accelerates, its traversal allocation $q_t/\Delta t$ increases while its compression q_c decreases. In the limit $v \to c$:

$$q_c \to 0, \qquad \frac{q_t}{\Delta t} \to q_{\text{stable}} \tag{13}$$

But $q_c = 0$ means the fold has no internal compression—no topological closure. It is no longer a stable fold; it is a photon. A massive particle reaching c would require complete conversion of its rest energy to traversal energy, at which point it ceases to be a particle.

The infinite energy requirement of special relativity [3] is reinterpreted: it is not that accelerating to c requires infinite energy input. It is that the fold's topological stability demands $q_c > 0$, and driving $q_c \to 0$ asymptotically requires unbounded energy reallocation because the fold resists dissolution.

4.5 Pair Production

The converse process is equally natural. When a photon's energy reaches $q_{\text{photon}} \geq q_{\text{stable}}$, the traversal quanta are sufficient to form a stable fold. The photon can convert to a particle–antiparticle pair—two stable folds of opposite topological orientation. The energy threshold for this process is:

$$E_{\text{pair}} = 2\, q_{\text{stable}} \cdot c^2 = 2m_0c^2 \tag{14}$$

This is the standard pair-production threshold [7, 6], here derived from the stability condition rather than from relativistic kinematics.

5 Falsifiable Predictions

The stability-budget framework generates predictions distinguishable from standard quantum electrodynamics:

1. **Substrate-derived Schwinger limit.** The maximum electric field strength before spontaneous pair production should be derivable from the phase-transition threshold q_{stable} and the substrate's constitutive properties $(\mu,\ \rho)$. If the derived value matches the QED Schwinger limit $E_S = m_e^2c^3/(e\hbar) \approx 1.3 \times 10^{18}$ V/m [6], this constitutes a nontrivial consistency check.

   Measurement context: Petawatt laser facilities (ELI, XCELS) approaching 10^{15} V/m; any deviation from QED pair-production rates at extreme fields would test the substrate prediction.

2. **Maximum single-photon amplitude.** The amplitude bound predicts a maximum electric field strength per photon, set by the phase-transition threshold. This is in principle measurable via single-photon field statistics in cavity QED experiments [8].

Measurement context: Superconducting microwave cavities with single-photon resolution; the predicted bound implies a saturation in field variance at the single-quantum level.

3. **Frequency-dependent Compton kinematics.** If the substrate's stability threshold varies with local conditions (near dense matter, strong gravitational fields), Compton scattering kinematics [5] should show environment-dependent corrections.

   Measurement context: Comparison of Compton scattering cross-sections measured at different gravitational potentials or near dense astrophysical objects. The Sunyaev–Zel'dovich effect [9] in galaxy clusters provides a natural laboratory where inverse Compton scattering occurs in varying gravitational environments.

4. **Photon frequency shift at phase boundaries.** Photons traversing sharp vacuum phase boundaries (e.g., near neutron star surfaces) should exhibit frequency perturbations distinct from gravitational redshift, arising from discontinuities in q_{stable}.

   Measurement context: Precision pulsar timing; comparison of frequency shifts with predictions from general relativity alone.

5. **Derivation of $\hbar$.** The identification of h as the energy cost per phase-boundary oscillation cycle predicts that $\hbar$ is expressible in terms of μ, ρ, and a phase-correlation length ℓ_0. A successful derivation would reduce the number of independent fundamental constants.

6 Open Frontiers

The stability-budget framework as presented here does not address:

- **Multi-photon absorption**: whether the velocity equation $v = c \cdot q_{\text{photon}}/q_{\text{stable}}$ is additive for sequential photon absorption, and how the budget handles oblique incidence.
- **Photon emission**: the reverse process, in which a particle sheds traversal quanta back into a propagating fold.
- **Quantum measurement**: the Born rule, wavefunction collapse, and entanglement remain outside the scope of this paper.
- **Action-principle derivation**: the stability condition (Eq. 1) is postulated here; its derivation from the substrate Lagrangian $\mathcal{L}(\mathbf{u}, \dot{\mathbf{u}}, \nabla\mathbf{u})$ is the subject of ongoing work.
- **Quadratic form proof**: as noted in Section 4.3, the orthogonality argument for the dispersion relation requires formal validation.

7 Discussion

The stability condition (Eq. 1) is a single algebraic relation governing the existence and kinematics of all substrate folds. From it, this paper derives or motivates:

- The photon's velocity c as an existence condition
- $E \propto f$ from oscillation mechanics
- Bounded amplitude and intensity as flux
- Polarization from compression geometry
- Resolution of wave–particle duality
- Momentum conservation as traversal-quanta accounting
- The massive-particle speed limit
- The pair-production threshold

The relativistic dispersion relation $E^2 = (pc)^2 + (m_0c^2)^2$ is argued to follow from orthogonality of the compression and traversal channels, but this argument awaits formal derivation (see Section 4.3).

None of these results require free parameters beyond the substrate's constitutive properties (μ, ρ, q_{stable}). The framework does not reproduce quantum mechanics in full—measurement

discreteness, entanglement, and the Born rule remain open frontiers. But the results presented here suggest that a significant portion of relativistic and quantum phenomenology may follow from a single mechanical balance condition on a phase-transitioning vacuum.

The stability equation occupies the same structural role in DPS that the metric tensor occupies in general relativity or the Hamiltonian occupies in quantum mechanics: it is the object from which dynamics are derived. Its full development—including connection to the substrate displacement field $\mathbf{u}(\mathbf{x}, t)$ and the action principle—is the subject of ongoing work.

The framework's relationship to existing programmes deserves comment. Analog gravity models [10, 11] demonstrate that effective spacetime geometries emerge from fluid and elastic media, but typically treat the medium as a background for wave propagation rather than as the ontological ground of particles themselves. Emergent spacetime approaches [12] work with superfluid condensates and recover aspects of the Standard Model at low energies. DPS differs in positing a *solid-like* substrate with phase transitions, which naturally produces topological defects (folds) as the particle sector and phase-transition propagation as the photon sector. Whether these approaches are complementary or competing remains an open question.

Data and Code Availability

No datasets were generated or analyzed in this study. All claims are derivable from Eq. (1) given the stated assumptions.

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References

[1] M. Planck, "Úber das Gesetz der Energieverteilung im Normalspectrum," *Ann. Phys.* **309**, 553–563 (1901).

[2] A. Einstein, "Úber einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt," *Ann. Phys.* **322**, 132–148 (1905).

[3] A. Einstein, "Zur Elektrodynamik bewegter Kórper," *Ann. Phys.* **322**, 891–921 (1905).

[4] P. A. M. Dirac, "The quantum theory of the emission and absorption of radiation," *Proc. R. Soc. Lond. A* **114**, 243–265 (1927).

[5] A. H. Compton, "A quantum theory of the scattering of X-rays by light elements," *Phys. Rev.* **21**, 483–502 (1923).

[6] J. Schwinger, "On gauge invariance and vacuum polarization," *Phys. Rev.* **82**, 664–679 (1951).

[7] C. D. Anderson, "The positive electron," *Phys. Rev.* **43**, 491–494 (1933).

[8] S. Haroche and J.-M. Raimond, *Exploring the Quantum: Atoms, Cavities, and Photons* (Oxford University Press, 2006).

[9] R. A. Sunyaev and Ya. B. Zel'dovich, "The observations of relic radiation as a test of the nature of X-ray radiation from the clusters of galaxies," *Comments Astrophys. Space Phys.* **4**, 173 (1972).

[10] C. Barceló, S. Liberati, and M. Visser, "Analogue gravity," *Living Rev. Relativ.* **8**, 12 (2005).

[11] W. G. Unruh, "Experimental black-hole evaporation?" *Phys. Rev. Lett.* **46**, 1351–1353 (1981).

[12] G. E. Volovik, *The Universe in a Helium Droplet* (Oxford University Press, 2003).

[13] T. Jensen, "Principia Substrata: A framework overview—emergent gravity, Lorentz

invariance, photon ontology, and dark matter from a phase-transitioning vacuum," Principia Academy (2026). https://principia.academy

[14] T. Jensen, "Gravitational lensing as anisotropic refraction in a hyper-elastic continuum," Zenodo (2026). https://doi.org/10.5281/zenodo.18736403

[15] T. Jensen, "Dynamic enforcement of Lorentz invariance via substrate stress feedback," Zenodo (2026). https://doi.org/10.5281/zenodo.18736487

[16] T. Jensen, "Shapiro delay as constitutive retardation in a hyper-elastic continuum," Zenodo (2026).

[17] T. Jensen, "Dark matter as energy-depleted space: rotation curves from substrate mode scarcity," Zenodo (2026). https://doi.org/10.5281/zenodo.18736625